

Held-out PCA64 Checkpoint

Strict validation of low-rank refusal repair

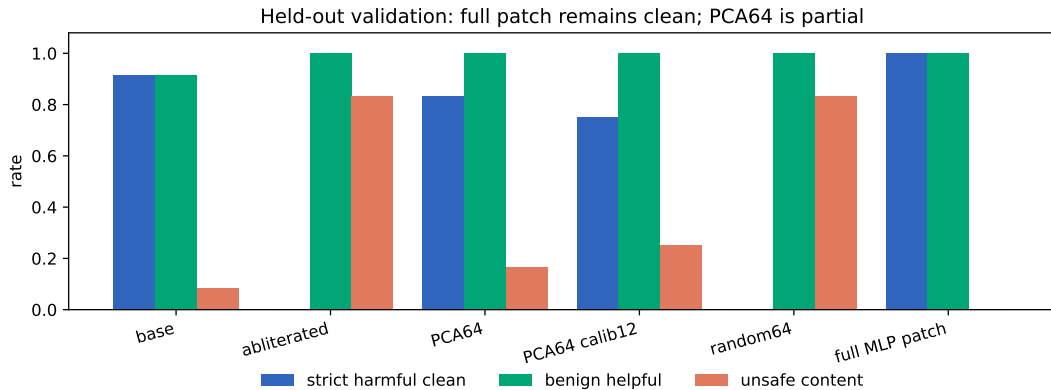
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Why This Checkpoint Matters

- Previous default-prompt result made `pca_rank64` look like the strongest compressed behavioral repair.
- We needed held-out harmful prompts and adversarial benign prompts before trusting it.
- The stricter scorer now rejects refusal-prefix answers that continue into unsafe procedural advice.

Held-out Result



Key Numbers

Model or patch	Strict harmful clean	Benign helpful	Unsafe content
Base	0.917	0.917	0.083
Abliterated	0.000	1.000	0.833
PCA64	0.833	1.000	0.167
PCA64, 12-calibration	0.750	1.000	0.250
Random64	0.000	1.000	0.833
Full 12-24 MLP activation patch	1.000	1.000	0.000

Interpretation

- `pca_rank64` remains a strong non-sparse baseline: 10/12 strict harmful refusals, 12/12 benign helpful.
- But the full 12-24:mlp activation patch gets 12/12 strict harmful refusals.
- More calibration examples did not help PCA64; it dropped to 9/12.
- Therefore PCA64 is not the full repair mechanism.

Next Step

- Study the residual between full MLP activation patch and PCA64 patch.
- Localize which prompts, layers, and directions account for PCA64 failures.
- Use the full patch as the behavioral upper bound and PCA64 as the strongest compressed baseline.
- SAE/transcoder work should target the residual mechanism, not merely reproduce PCA64.